Operations Research for Health Care 44 (2025) 100453

Contents lists available at ScienceDirect

Operations Research for Health Care

journal homepage: www.elsevier.com/locate/orhc

Predicting leishmaniasis outbreaks in Brazil using machine learning models based on disease surveillance and meteorological data

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ARTICLE INFO

Dataset link: https://github.com/AndreCintas/leishmaniasis

Keywords:
Leishmaniasis
Outbreak prediction
Machine learning
Artificial neural network
Support vector machine

ABSTRACT

Leishmaniasis poses a significant global health concern due to the absence of vaccines for humans and high infection rates in some countries. It is classified as a neglected tropical disease. In 2022, roughly 85% of global visceral leishmaniasis cases were reported in seven countries: Brazil, Ethiopia, India, Kenya, Somalia, South Sudan, and Sudan. Despite Brazil's advanced medical capabilities compared to other affected regions, certain areas still witness a significant number of cases, prompting increased attention from researchers and raising concerns within the healthcare system. This study explores the application of artificial intelligence algorithms, particularly machine learning (ML) models to predict leishmaniasis outbreaks in selected Brazilian cities based on accumulated cases from 2007 to 2022, leveraging available meteorological data to enhance model accuracy. Our investigation concentrated on the following cities in Brazil: Fortaleza/CE, Teresina/PI, and São Luís/MA were chosen for the study of visceral leishmaniasis, whereas Manaus/AM, Rio Branco/AC, and Macapá/AP were selected for the study of tegumentary leishmaniasis, encompassing both cutaneous and mucocutaneous forms. Several Artificial Neural Network (ANN) architectures were evaluated, including a Simple Feedforward Neural Network (SFNN), a Deep Feedforward Neural Network (DFNN), and a Long Short-Term Memory (LSTM) recurrent neural network. Additionally, the Support Vector Machine (SVM), specifically the Support Vector Regression (SVR), was tested. Various metrics were used to identify the most effective models, in which the Root Mean Squared Error (RMSE) was the primary one. The results highlight the significance of meteorological data as a crucial factor in ML models for predicting leishmaniasis outbreaks, while also emphasizing the importance of fine-tuning these models to achieve greater accuracy. Finally, data and the pseudo-code of the models are accessible through an open repository to encourage further studies in this area.

1. Introduction

Leishmaniasis is an infectious disease caused by parasitic protozoa of the genus *Leishmania*, comprising approximately fifty species globally, of which twenty species are pathogenic to humans [1]. Transmission occurs through the bite of an infected insect, known as a phlebotomine and commonly referred to as a sandfly. This insect belongs to the subfamily of the family *Psychodidae*, and only the female exhibits this feeding habit [2]. The larvae of these insects mature in damp soil and organic debris, distinguishing them from common mosquitoes that breed in water [3].

In humans, leishmaniasis manifests in several forms, and the most severe is visceral leishmaniasis (VL), also known as kala-azar. This condition affects internal organs such as the spleen, liver, and bone marrow, causing their enlargement. The most common form is cutaneous leishmaniasis (CL), which results in skin lesions. Another form is mucocutaneous leishmaniasis, characterized by gradually worsening ulcerations of the mucosa, affecting areas such as the mouth, nose, and throat [2,4,5]. In Brazil, the latter two forms are known to be part of American tegumentary leishmaniasis [6], here simply denoted as tegumentary leishmaniasis (TL).

According to the World Health Organization (WHO), leishmaniasis is classified as a neglected tropical disease [7]. As reported by WHO, in 2022, approximately 85% of global visceral leishmaniasis cases were concentrated in seven countries: Brazil, Ethiopia, India, Kenya, Somalia, South Sudan, and Sudan [8]. Moreover, roughly 85% of cutaneous leishmaniasis cases occurred in eight countries: Afghanistan, Algeria, Brazil, Colombia, Iran (Islamic Republic of), Iraq, Peru, and the Syrian Arab Republic.

In the Americas, Brazil emerges as the leading endemic country for VL [2], representing 90% of cases in Latin America [9]. While 80% of

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https://doi.org/10.1016/j.orhc.2024.100453
Received 23 April 2024; Received in revised form 21 October 2024; Accepted 11 November 2024
Available online 20 November 2024
2211-6923/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

CL cases are reported in the Eastern Mediterranean region [2], Brazil has consistently reported an average of 22,000 cases per year of TL from 2007 to 2022, as revealed by exploratory data analysis. Both types of leishmaniasis are mandatory notifiable diseases within the Health System in Brazil [10].

The highest risk of infection is likely associated with factors such as socioeconomic vulnerability and climate conditions conducive to mosquito breeding. Poverty exacerbates the risk of infection due to inadequate housing and sanitation, which can promote the proliferation of sandflies. Variations in temperature and precipitation can impact mosquito population sizes and geographical spread. Additionally, living in or traveling to endemic areas, increased susceptibility among malnourished individuals, and environmental changes such as deforestation and unplanned urbanization are also contributing factors [2].

In Brazil, the media often reports the significant strain on healthcare systems, where appointment scheduling can stretch for months, and medical staff shortages persist, among other issues, raising concerns about potential harm to individuals [11]. In this scenario, the capability of algorithms to analyze data holds considerable promise, aiding professionals in operations resource planning and disease diagnosis [12]. Such assistance offers valuable guidance for more precise treatments and accelerates the delivery of medical care.

Another approach involves predicting (e.g., by analytical models) disease outbreaks and transitioning from individual patient treatment to concentrating on the community or specific regions of interest. This entails analyzing disease surveillance data alongside other relevant information, such as meteorological and socioeconomic data.

Brazil occupies a pivotal role in the realm of leishmaniasis, given the significant incidence of cases in certain areas, thus rendering the current study particularly pertinent. Our objective is to bolster proactive measures to contain and manage the spread of the disease, thereby equipping authorities and governments with the tools to formulate effective strategies. These strategies would mitigate adverse effects and facilitate the optimal allocation of resources for both prevention and patient treatment.

To provide some context, Carvalho et al. [13] reported that the Health System in Brazil required an amount of approximately one million dollars in 2014 to assist patients with VL. It is likely that this number is significantly higher at present.

This study aimed to investigate machine learning models for predicting the prevalence of leishmaniasis cases in cities with elevated disease incidence. It utilizes various data inputs, including the number of reported leishmaniasis cases from previous months, compensated mean air temperature, insolation, precipitation, and relative humidity. The choice of our models was inspired by the study of Roster and Rodrigues [14] applied to dengue cases, which examined 19 studies using ANN.

Although time series models are commonly used for predicting leishmaniasis, as will be detailed in the literature review, our approach diverges by incorporating a range of artificial intelligence algorithms, including the Simple Feedforward Neural Network, the Deep Feedforward Neural Network, the Long Short-Term Memory recurrent neural network, and the Support Vector Regression. These models leverage the existing computational advancements that are readily accessible. Their development relies on a combination of leishmaniasis cases and meteorological data. Additionally, supplementary analyses of the socioeconomic data of the cities were undertaken to further elucidate the relationship between the disease and these data.

The article is structured as follows. Section Literature review focuses on machine learning models and commonly used variables for predicting infectious disease outbreaks. Section Materials and Methods provides an explanation of the study regions and the data utilized. An exploratory data analysis is conducted to better understand leishmaniasis in Brazil, followed by a detailed description of the entire machine learning modeling process. Section Results and Discussions presents the selected models for generating the outcomes, as well as a discussion of the tested predictor variable combinations and the evaluation metrics. Finally, Section Concluding Remarks summarizes the key findings and proposes further studies in this line of research.

2. Literature review

The literature review was initially conducted to evaluate the utilization of ML models in addressing infectious diseases. Extensive research exists on diseases such as dengue, a mosquito-borne viral infection transmitted by *Aedes spp* [15]. In such studies, it is customary to investigate correlations with meteorological data as, e.g., in Aleixo et al. [16], Jain et al. [15], Momoh and Fügenschuh [17], Ramadona et al. [18], Chumpu et al. [19], Rocha et al. [20] and Herath et al. [21]. Furthermore, diseases such as malaria caused by the protozoa *Plasmodium spp* and transmitted by female *Anopheles* mosquitoes are also examined within this context, often incorporating meteorological data as, e.g., in Haddawy et al. [22] and Gomez-Elipe et al. [23].

Both dengue and malaria outbreaks have been extensively explored using ML models. During the literature search, a significant amount of research on leishmaniasis was found, but all focused on applying the traditional time series models, as observed by Li et al. [24], Pimentel et al. [25] and Sharafi et al. [26]. Given the lack of ML studies in the context of leishmaniasis, we employ modern artificial intelligence techniques, such as ANN and SVM algorithms in our study.

In the study conducted by Aleixo et al. [16], the total number of reported dengue cases were sourced from the Brazilian National System of Information on Notifiable Diseases (SINAN), while meteorological data (precipitation, air temperature, and humidity) were obtained from the Brazilian National Weather Institute (INMET), and socio-demographic data were collected from the Brazilian Institute of Geography and Statistics (IBGE) [27]. These data were utilized in a CatBoost model to predict dengue outbreaks in the city of Rio de Janeiro from 1 to 3 months in advance. The CatBoost model exhibited significant improvements over the commonly employed Seasonal Autoregressive Integrated Moving Average (SARIMA) model.

The study by Jain et al. [15] focused on predicting dengue outbreaks across fifty districts in Thailand using a ML-based methodology, specifically Generalized Additive Models (GAM). The data utilized encompass meteorological, clinical, disease surveillance, socio-economic factors, and spatial dependence variables. Evaluation metrics include RMSE, Standardized RMSE (SRMSE), Adjusted R-squared, Deviance explained, and change in Akaike Information Criterion (AIC). The study evaluated six different models, with the most effective one incorporating lagged meteorological data, lagged dengue data from target districts and their surroundings, and socioeconomic data for one-month lead time forecasts.

In the city of Yogyakarta, Indonesia, Ramadona et al. [18] investigated the prediction of dengue outbreaks using three strategies that involved combining variables in the models: utilizing only meteorological variables, only time-lagged disease surveillance variables, and a combination of both. Generalized Linear regression models (GLM) were applied to uncover the relationship between predictor variables and dengue surveillance data spanning from 2001 to 2010, with external validation conducted on data from 2011 to 2013. Model evaluation focused on epidemic detection and predicted case numbers using metrics such as RMSE, SRMSE, and AIC. The optimal model, incorporating both surveillance and meteorological data, demonstrated the ability to detect outbreaks up to two months in advance.

Chumpu et al. [19] conducted a study to predict dengue outbreaks in 65 out of 77 provinces in Thailand using data from 2001 to 2014. It incorporated eight meteorological variables, such as precipitation, temperature, humidity, wind direction, and wind power, with lag times ranging from one to three weeks, alongside lagged dengue case variables spanning one to three weeks. Several statistical regression models, including Poisson regression, negative binomial regression, quasi-likelihood regression, as well as time-series models such as Autoregressive Integrated Moving Average (ARIMA) and its seasonal counterpart (SARIMA) were employed. Results indicated that the one-week lagged dengue cases were the most influential predictor across most provinces, followed by precipitation and wind-related variables.

The study concluded that prediction models should be constructed at the provincial level due to the unique geographical and meteorological factors influencing dengue dynamics.

Dengue cases pose a significant challenge to Sri Lanka's healthcare system. In a study conducted by Herath et al. [21], an ANN was developed to predict dengue outbreaks. The model incorporates seven weather variables, such as average temperature, relative humidity, total rainfall, rainy days per week, with lagged weeks and previous cases as input parameters. Several combinations of neurons and activation functions were explored, and the configuration with four neurons in the hidden layer achieved the best performance using a MLP network.

The study carried out by Roster and Rodrigues [14] aimed to identify the artificial neural networks applied in predicting dengue cases. Analyzing nineteen papers, the study revealed that most of them relied on historical dengue cases and meteorological data. Several types of neural networks were used, including ANN with one hidden layer, deeper feedforward neural networks, recurrent neural networks such as LSTM and Gated Recurrent Unit, and Convolutional Neural Network. The performance of these models was evaluated, often compared to the SVM.

Similar to dengue, various models were used in forecasting malaria outbreaks. Gomez-Elipe et al. [23] employed ARIMA time series models, to forecast malaria cases one month ahead in the Karuzi region of Burundi. The study incorporated environmental factors such as the normalized difference vegetation index (NDVI), mean maximum temperature, and rainfall, along with the number of malaria cases in the preceding month. The model achieved a forecasting accuracy of 93% within a margin of ± 4 cases per 100 inhabitants, highlighting its potential as a straightforward yet effective tool for predicting malaria. Another approach was taken by Haddawy et al. [22], who employed Bayesian Networks to predict malaria cases in the Tha Song Yang district in northern Thailand. The study integrated disease data with environmental variables, capturing time-lagged and nonlinear relationships. The model demonstrated high accuracy for one and two week predictions at a village-level spatial resolution.

In addition to this review on dengue and malaria, our investigation sought analytical models used for predicting leishmaniasis outbreaks. Li et al. [24] adopted a time series model, ARIMA and hybrid ARIMA-EGARCH model, to forecast VL outbreaks in Kashgar, located on the southern border of Xinjiafng, China, using AIC measures, Bayesiano Information Criterion (BIC) and Mean Absolute Percentage Error (MAPE) as metrics. The ARIMA application did not yield satisfactory results and the exploration of the ARIMA-EGARCH hybrid model demonstrated improved predictive capabilities.

Similarly, Pimentel et al. [25] employed the SARIMA model to forecast VL outbreaks in the state of Maranhão, Brazil. Using time series data from 2001 to 2018, the study analyzed 10,431 reported cases, with a higher incidence among children aged 0 to 14 years and a predominance of male cases (61.71%), the study concluded that SARIMA is a valuable tool for predicting VL incidence and guiding health intervention measures.

Sharafi et al. [26] adopted the SARIMA model to forecast CL in the southern region of Fars province, Iran, using data from January 2010 to March 2016. The study also found that temperature with a three-month lag increased disease incidence, rainfall with a four-month lag decreased it, and rainfall with a nine-month lag increased it. The results highlighted that CL follows a seasonal pattern and that prolonged drought conditions are associated with a declining trend in the disease.

Table 1 provides a summary of key studies, situating our research in the context of this domain. While the employment of ML models has gained prominence across several areas of healthcare, such as medical diagnosis and epidemiology, its direct application to predicting leishmaniasis outbreaks remains limited, as observed. Developing ML-based strategies to predict and respond to leishmaniasis outbreaks holds considerable promise, offering the potential to enhance the efficacy of control measures and mitigate the impact of this disease on affected communities, especially in a vast country such as Brazil.

3. Materials and methods

3.1. Brazilian districts

This study considers the entire Brazilian territory, selecting specific cities based on the highest number of reported cases from 2007 to 2022 in the SINAN database [28]. The chosen cities also had meteorological data available in the INMET database [29], aligning with the study's objective to incorporate meteorological data into the models. For the analysis of VL, the selected cities were Fortaleza in the state of Ceará (CE), Teresina in the state of Piauí (PI), and São Luís in the state of Maranhão (MA), all situated in the Northeast region of the country. For TL, the chosen cities were Manaus in the state of Amazonas (AM), Rio Branco in the state of Acre (AC), and Macapá in the state of Amapá (AP), all located in the Northern region.

The cities were classified based on the total number of cases accumulated from 2007 to 2022. In terms of VL, Fortaleza ranked first with 3754 cases, followed by Teresina, with 3674 cases in second place, and São Luís, with 2782 cases in third place. Notably, meteorological data are readily accessible for all these cities.

Similarly, the cities affected by TL were ranked based on the number of cases. Presidente Tancredo Neves, located in the state of Bahia (BA), claimed the top position with 10,743 cases, followed by Manaus with 10,730 cases in the second position, Rio Branco with 4019 cases in the third position, Taperoá, located in the state of Paraíba (PB), in the fourth position with 3328 cases, Ituberá, also in the state of Bahia, in the fifth position with 2868 cases, and Macapá in the sixth position with 2825 cases. The cities of Presidente Tancredo Neves/BA (first), Taperoá/PB (fourth), and Ituberá/BA (fifth) were not included in our study due to the unavailability of meteorological data, which is necessary to apply the models.

3.2. Demographic and socioeconomic profile

In , demographic, socioeconomic, and urban development data are provided to enhance comprehension of the selected cities in the study. Each line of this table contains:

- Population: the population size of the city.
- Sanitary Sewage: percentage of access to basic sanitation.
- Territorial Area: territorial extension of the city in square kilometers (km^2).
- Urbanized Area: the area of the city that is urbanized (km^2).
- MHDI: the Municipal Human Development Index is a comprehensive indicator that considers several socioeconomic factors, such as income, education, and health, to assess the overall quality of life in a given region. A higher MHDI value indicates a higher level of socioeconomic development in the area [30].
- Monthly Wage: average monthly salary of formal workers, quantified in terms of minimum wages.
- Employed Population: the number of people employed in the city.
- Low Income: percentage of the population with nominal monthly per capita income of up to half the minimum wage, indicating economic inequalities.
- School Attendance: schooling rate from 6 to 14 years of age, indicating access to elementary education.
- GDP per capita: the average economic production per person in the city.
- Child Mortality: number of deaths per thousand live births, reflecting the quality of healthcare.

The data shows that Macapá has the lowest percentage of basic sanitation (26.8%), the highest infant mortality rate (19.65%), and the lowest GDP. Rio Branco records the lowest MHDI at 0.727 and the second-lowest basic sanitation rate (56.7%). Manaus boasts the highest GDP among the selected cities due to a large industrial zone

Table 1
Relevant literature for this study's context.

Authors	Disease[a]	Country	Geospatial scale	Input data	Data aggregation	Model	Evaluation metric
Aleixo et al. [16]	Dengue	Brazil	Districts	Epidemiological, Meteorological, Demography, Number of Public Health Facilities	Monthly	CatBoost, SARIMA	R^2, MAE, MAPE, RMSE
Jain et al. [15]	Dengue	Thailand	Districts	Epidemiological, Meteorological, socio-economic, Clinical	Monthly	GAM	R^2, RMSE, SRMSE, AIC
Haddawy et al. [22]	Malaria	Thailand	District	Epidemiological, Meteorological, Satellite	Weekly	Bayesian Networks	MAE
Ramadona et al. [18]	Dengue	Indonesia	District	Epidemiological, Meteorological	Monthly	GLM	RMSE, SRMSE, AIC
Chumpu et al. [19]	Dengue	Thailand	Provinces	Epidemiological, Meteorological	Weekly	Regression, (S)ARIMA	R^2, RMSE
Gomez-Elipe et al. [23]	Malaria	Burundi	Province	Epidemiological, Environmental	Monthly	ARIMA	Adjusted R^2
Li et al. [24]	VL	China	District	Epidemiological	Monthly	ARIMA, ARIMA-EGARCH	RMSE, AIC, BIC
Pimentel et al. [25]	VL	Brazil	State	Epidemiological	Monthly	SARIMA	MAPE, AIC, BIC
Sharafi et al. [26]	CL	Iran	Province	Epidemiological, Meteorological	Monthly	SARIMA	AIC, BIC
Herath et al. [21]	Dengue	Sri Lanka	Country	Epidemiological, Meteorological	Weekly	Neural Networks	MMRE, Pred(m)
This study	**VL/TL**	**Brazil**	**Districts**	**Epidemiological, Meteorological**	**Monthly**	**Neural Networks SVR**	**RMSE, MSE, MAE**

[a] VL – Visceral Leishmaniasis, TL – Tegumentary Leishmaniasis, CL – Cutaneous Leishmaniasis.

Table 2
Socioeconomic and demographic indicators.

City/state	Population 2022	Sanitary sewage 2010 (%)	Territorial area 2022 (km^2)	Urbanized area 2019 (km^2)	MHDI 2010
Fortaleza/CE [31]	2,428,708	74	312,353	253,69	0.754
Teresina/PI [32]	866,300	61.6	1,391,293	172,26	0.751
São Luís/MA [33]	1,037,775	65.4	583,063	165,96	0.768
Manaus/AM [34]	2,063,689	62.4	11,401,092	277,09	0.737
Rio Branco/AC [35]	364,756	56.7	8,835,154	87,42	0.727
Macapá/AP [36]	442,933	26.8	6,563,849	82,29	0.733

City/State	Monthly wage 2021	Employed population 2021	Low income 2010 (%)	School attendance 2010 (%)	GDP per capita 2020 (R$)	Child mortality 2020 (‰)
Fortaleza/CE [31]	2.7	848,283	36.9	96.1	24,253.93	12.02
Teresina/PI [32]	2.6	289,423	38.6	97.8	24,858.31	12.1
São Luís/MA [33]	3.1	389,518	38.8	96.8	29,823.95	13.09
Manaus/AM [34]	3.0	554,913	37.9	94.2	41,345.11	12.8
Rio Branco/AC [35]	3.3	106,966	36.4	95.1	23,171.69	14.97
Macapá/AP [36]	3.8	106,360	39.0	94.8	22,880.7	19.65

with subsidized taxes. São Luís has the highest MHDI (0.768) among the selected cities. Teresina reports the lowest average monthly salary (2.6 of the minimum wage), and Fortaleza records the highest basic sanitation rate (75%). Getulio Vargas Foundation [37] (FGV) assesses the level and evolution of poverty in Brazil through the New Poverty Map [38]. The study classifies Brazil into 146 geographic strata, where Number 1 represents the highest poverty index and 146 the lowest. In 2021, the positions of the selected cities were as follows below.

- Fortaleza: 73rd position with 30.75%;
- Teresina: 76th position with 30.33%;
- São Luís: 58th position with 37.86%;
- Manaus: 56th position with 38.93%;
- Rio Branco: 57th position with 38.29%;
- Macapá: 72nd position with 32%.

The proximity in rankings of Macapá, Fortaleza, and Teresina, as well as São Luís, Manaus, and Rio Branco, on FGV Social's New Poverty Map, implies comparable levels of poverty, measured by the percentage of individuals with a per capita family income of up to 497 Brazilian reais per month (i.e., around 100 USD, in the exchange rate of January 2024).

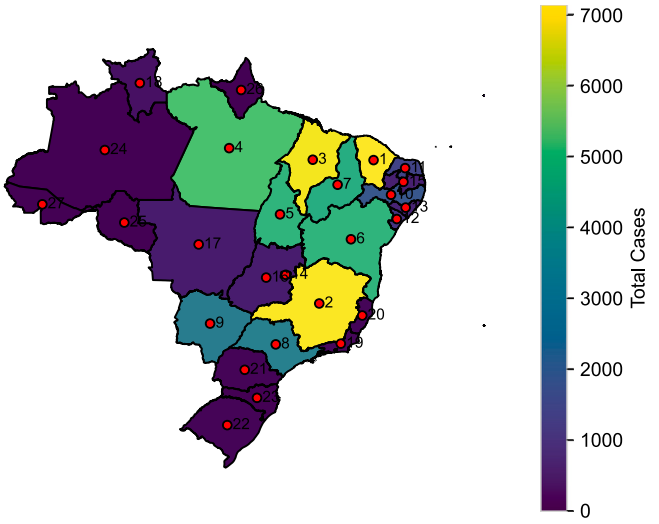


Fig. 1. Accumulated cases of visceral leishmaniasis in Brazil from 2007 to 2022. The numbers inside each state are associated with Table 3.

Table 3
Reported cases: ranking of visceral leishmaniasis [28].

Position	State	Cases	Share of the total (%)
1	Ceará	7133	13.18%
2	Minas Gerais	7091	13.10%
3	Maranhão	7021	12.97%
4	Pará	5055	9.35%
5	Tocantins	4650	8.59%
6	Bahia	4629	8.56%
7	Piauí	4405	8.15%
8	São Paulo	3075	5.68%
9	Mato Grosso do Sul	2975	5.50%
10	Pernambuco	1934	3.57%
11	Rio Grande do Norte	1408	2.60%
12	Sergipe	1007	1.86%
13	Alagoas	709	1.31%
14	Distrito Federal	640	1.18%
15	Paraíba	619	1.14%
16	Goiás	546	1.01%
17	Mato Grosso	515	0.95%
18	Roraima	284	0.52%
19	Rio de Janeiro	127	0.23%
20	Espírito Santo	104	0.19%
21	Paraná	69	0.13%
22	Rio Grande do Sul	63	0.12%
23	Santa Catarina	22	0.04%
24	Amazonas	19	0.04%
25	Rondônia	9	0.02%
26	Amapá	6	0.01%
27	Acre	–	–

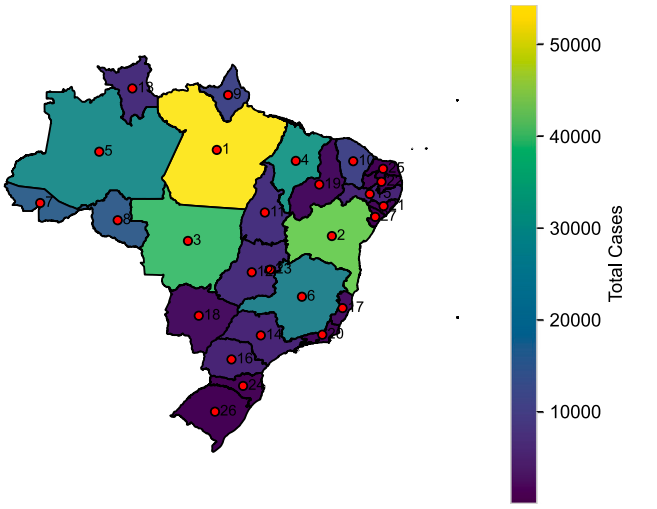


Fig. 2. Accumulated cases of tegumentary leishmaniasis in Brazil from 2007 to 2022. The numbers inside each state are associated with Table 4.

Table 4
Reported cases: ranking of tegumentary leishmaniasis [28].

Position	State	Cases	Share of the total (%)
1	Pará	54 238	17.00%
2	Bahia	42 309	13.26%
3	Mato Grosso	37 827	11.85%
4	Maranhão	29 091	9.12%
5	Amazonas	26 692	8.36%
6	Minas Gerais	24 240	7.59%
7	Acre	16 975	5.33%
8	Rondônia	16 252	5.11%
9	Amapá	11 215	3.52%
10	Ceará	10 978	3.45%
11	Tocantins	7566	2.37%
12	Goiás	7055	2.22%
13	Roraima	6959	2.19%
14	São Paulo	5716	1.80%
15	Pernambuco	5636	1.77%
16	Paraná	5441	1.71%
17	Espírito Santo	2194	0.69%
18	Mato Grosso do Sul	1947	0.61%
19	Piauí	1664	0.52%
20	Rio de Janeiro	1066	0.33%
21	Alagoas	1056	0.33%
22	Paraíba	981	0.31%
23	Distrito Federal	847	0.27%
24	Santa Catarina	503	0.16%
25	Rio Grande do Norte	250	0.07%
26	Rio Grande do Sul	146	0.04%
27	Sergipe	120	0.03%

3.3. Exploratory data analysis

Figs. 1 and 2 illustrate the accumulated leishmaniasis cases in Brazilian states from 2007 to 2022 for VL and TL, respectively. The numbers identify each state by ranking them in descending order according to the number of cases. Notably, the highest occurrences of VL are observed in the states of Ceará, Minas Gerais, and Maranhão. Conversely, for TL, the state of Pará stands out among all the others. In addition to the map highlighting states with the highest occurrences, a comprehensive ranking of all states and the federal district is presented in Table 3 for VL and Table 4 for TL [28]. A dash indicates the state of Acre, where no data on VL is available. These tables also contain the numbers identifying the states in each figure.

The total number of VL cases from 2007 to 2022 is 54,115. The northeastern region showed the highest percentage, accounting for 53.34% of the cases, followed by the southeastern region with 19.21%, the northern region with 18.52%, the central-west region with 8.64%, and lastly, the southern region with 0.29%.

Regarding VL, the disease is not uniformly spread in the country, the states of Ceará and Minas Gerais represent approximately 13.1% of the cases each, followed by Maranhão with 12.9%, Pará with 9.3%, and Tocantins with 8.5%.

Delving deeper into VL at the city level, our objective was to pinpoint urban areas with the highest disease incidence for the application of ML models. Over the period spanning from 2007 to 2022, a total of 1709 cities reported cases. Among them is Fortaleza with 6.9% of the cases, followed by Teresina with 6.7%, São Luís with 5.1%, Belo Horizonte with 4.9%, and Araguaína with 4.3%.

The same data exploration is undertaken for TL, considering the period from 2007 to 2022, which recorded a total of 318,964 cases in Brazil. This confirms the higher prevalence of TL compared to the VL form. Similarly to the prominence of VL in the northeastern region, the northern region takes the lead in TL, with 43.86% of cases,

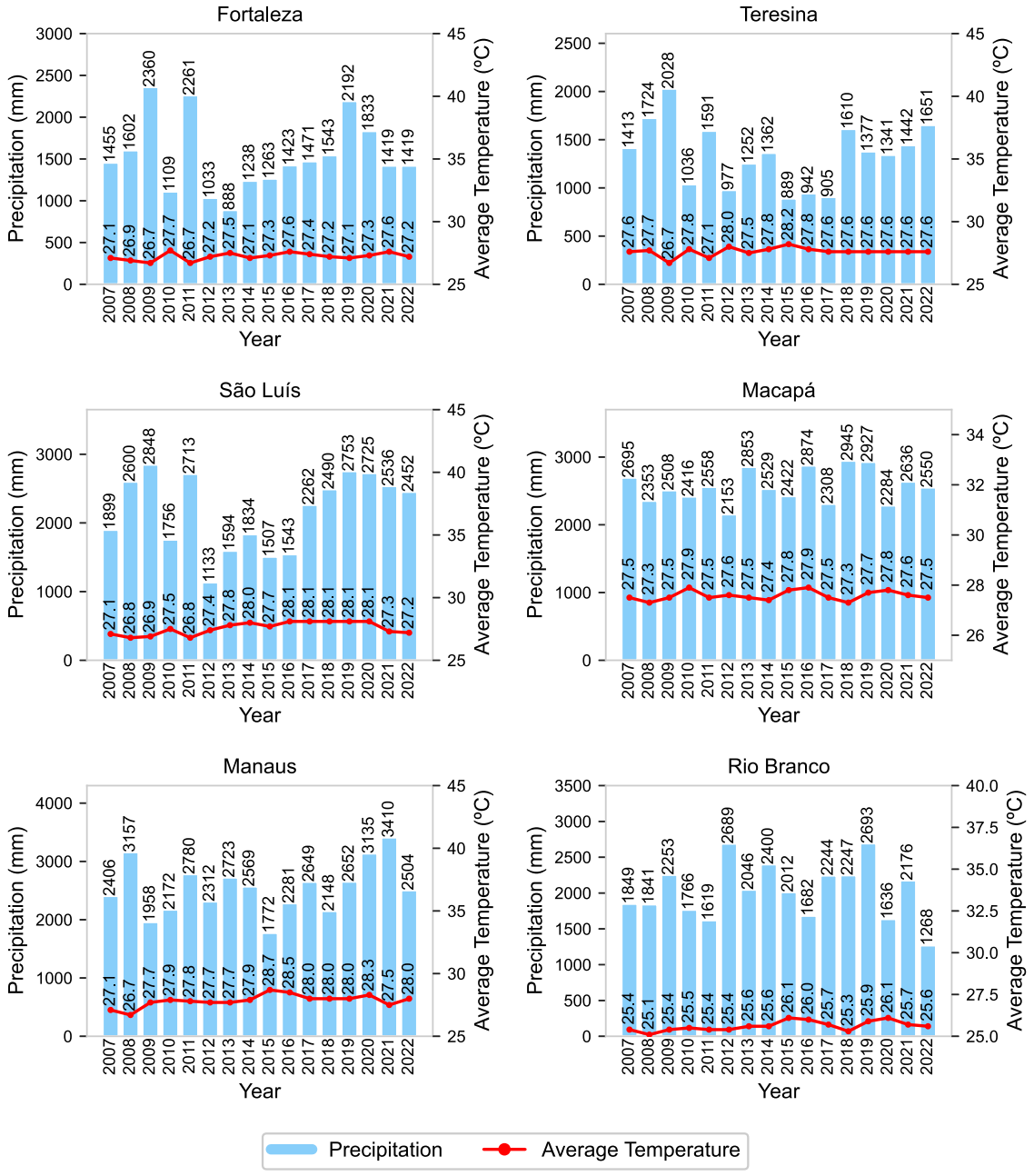


Fig. 3. Accumulated precipitation and average air temperature of the cities under investigation [29].

followed by the northeastern region with 28.87%, central-west with 14.95%, southeastern with 10.41%, and southern with 1.91%. The low incidence of cases in the southern region could be understood as the disease not being endemic in this area.

The state of Pará leads with 17.0% of disease occurrence, followed by Bahia with 13.2%, Mato Grosso with 11.8%, Maranhão with 9.1%, and Amazonas with 8.3%. At the city level, a total of 3612 cities reported TL cases in Brazil; where the highest occurrences were in Presidente Tancredo Neves/BA (3.3%), Manaus/AM (3.3%), Rio Branco/AC (1.2%), Taperoá/PB (1.0%), Ituberá/BA (0.8%), and Macapá/AP (0.8%).

In the SINAN database [28], monthly case numbers are available and categorized into visceral and tegumentary (cutaneous and mucocutaneous) forms. Data from INMET [29] includes total insolation (hours), total precipitation (millimeters), compensated mean air temperature (Celsius degrees), and relative humidity (%). In Fig. 3, annual data on accumulated precipitation and average air temperature are presented [29]. Analyzing the graphs, we observe that the average annual temperatures remain relatively stable, with significant variations in accumulated precipitation.

The epidemiological overview of VL and TL from 2007 to 2022 in Brazil is illustrated in Fig. 4 [28]. Despite a reduction in VL cases in recent years, specifically from 2018, the data show an increase in lethality, as depicted in Fig. 5 [28]. This increase could be attributed to improved case notification, not lethality itself. On the other hand, in the TL form, the average reported cases surpass that of VL by more than sixfold, although with significantly lower lethality. This analysis enables a comprehensive understanding of the geographical spread of the disease across Brazil, providing insights, e.g., relative to lethality.

The data indicate that the VL form is much more lethal than the TL form. Despite its low lethality, the TL form causes damage throughout the individual's life, requiring frequent medical care. This potentially weakens and reduces the overall quality of life and sometimes reduces the ability to work.

For VL, the southern region comprises only 0.29% of the total number of cases but demonstrates the highest lethality rate at 10.3%,

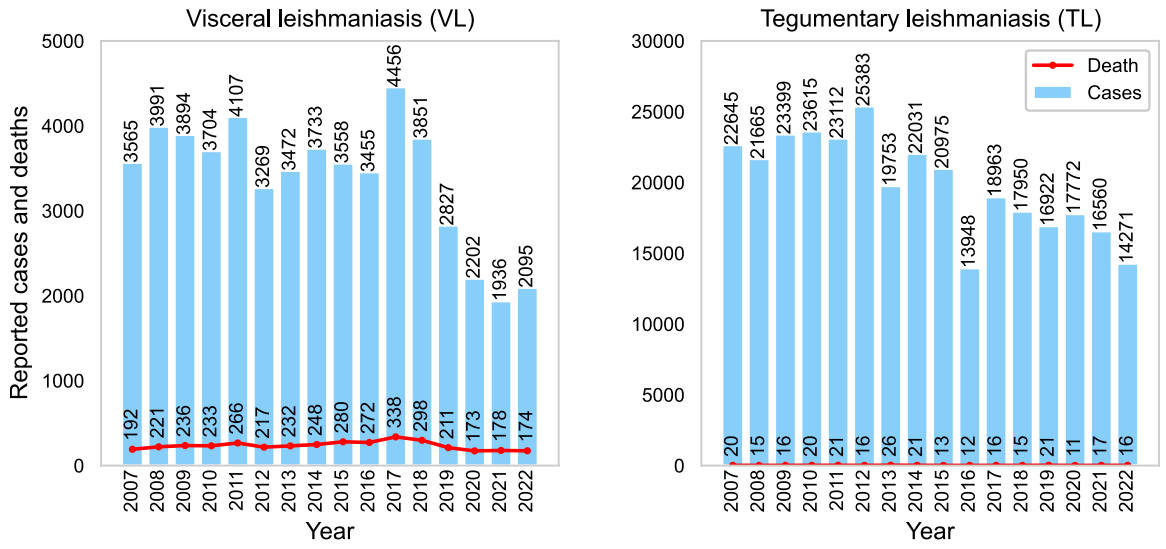


Fig. 4. Leishmaniasis cases and deaths in Brazil from 2007 to 2022 [28].

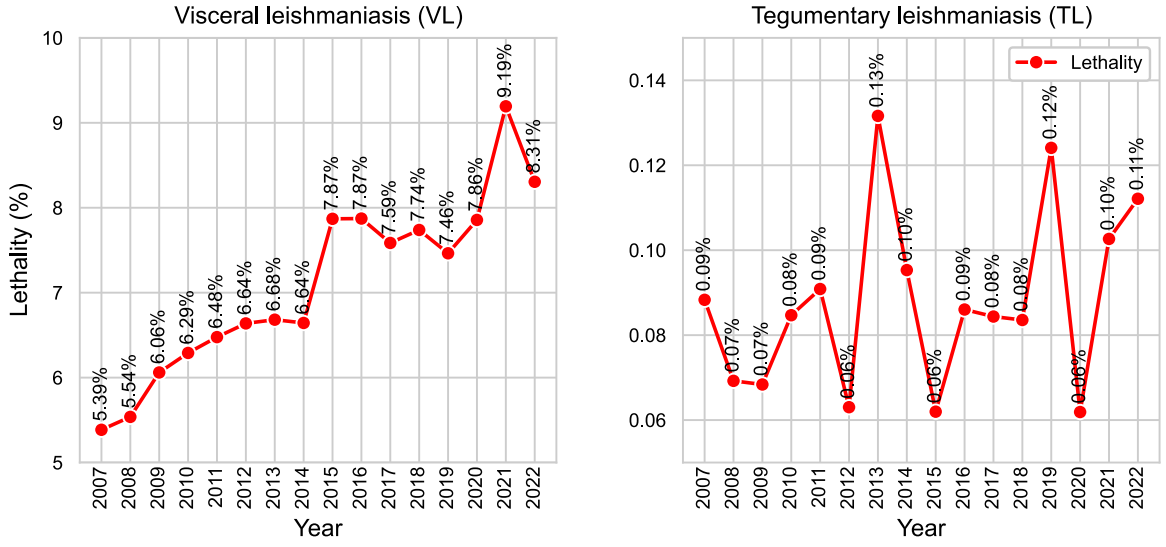


Fig. 5. Leishmaniasis lethality in Brazil from 2007 to 2022 [28].

but this may be due to the death of the few people in the statistics. Following closely are the southeastern region at 9.0%, central-west at 7.7%, northeastern at 6.8%, and northern at 4.8% lethality. Notably, regions with higher incidence rates do not necessarily exhibit the highest lethality possibly due to notification discrepancies. The states with the highest death rates were Amazonas (15.7%), Rio de Janeiro (14.1%), Paraná (13.0%), Rondônia (11.1%), and Rio Grande do Sul (11.1%). Some states with lower case numbers, such as Amapá and Santa Catarina, report no deaths attributed to VL.

Applying the same approach to assess lethality at the city level for VL reveals rates nearing 10% in Aracaju, Belo Horizonte, Palmas, Imperatriz, and Maceió. However, for the cities selected in this study, Fortaleza exhibits a lethality rate of 6.9%. Teresina and São Luís show rates of 6.9% and 7.5%, respectively.

Besides that, an analysis of age distribution was conducted, splitting ages into eleven categories: under 1 year, 1–4 years, 5–9 years, 10–14 years, 15–19 years, 20–39 years, 40–59 years, 60–64 years, 65–69 years, 70–79 years, and 80 years or older. The highest incidences of VL are observed in the age groups of 1–4 years (24.5%) and 20–39 years (22.6%).

In terms of TL lethality, the values are notably lower compared to VL, as observed in Fig. 5. The southern region records the highest rate, although still low at 0.3%. This is followed by the southeastern region at 0.2%, central-west at 0.09%, northeastern at 0.08%, and northern at 0.04%. Despite the northern and northeastern regions having more cases, constituting 72.73% of the total, they exhibit the lowest lethality rates at 0.04% and 0.08%, respectively.

The only state with a lethality exceeding 1% is Rio Grande do Sul, standing at 1.3%, while all others remain below 1%. Given that numerous cities reported cases, including some with only 1 or 2 cases, lethality was filtered to cities with reported case numbers above 500. This was done to prevent situations where a city with just one infected patient and subsequent death would indicate a 100% lethality. As a result, only Palmas reported a lethality exceeding 1%, more precisely with 1.2%, and all other cities showed rates below 1%. For the cities selected in this study, Manaus and Macapá do not report deaths due to TL, and Rio Branco has a 0.04% lethality rate. The age group with the highest incidence of TL is 20–39, with a value of 39.3%, followed by the 40–59 age group, which accounts for 24.5%.

3.4. Machine learning approach

The construction of machine learning-based models needs essential steps to improve accuracy and achieve superior results. This study

Table 5
Missing values and outliers per city and variable.

City	Insolation		Total precipitation		Average air temperature		Air humidity		Leishmaniasis	
	Missing	Outliers	Missing	Outliers	Missing	Outliers	Missing	Outliers	Missing	Outliers
Fortaleza	28 (14.58%)	0 (0.00%)	23 (11.98%)	2 (1.04%)	11 (5.73%)	0 (0.00%)	12 (6.25%)	0 (0.00%)	0 (0.00%)	4 (2.08%)
Teresina	43 (22.40%)	0 (0.00%)	16 (8.33%)	3 (1.56%)	90 (46.88%)	0 (0.00%)	100 (52.08%)	0 (0.00%)	0 (0.00%)	2 (1.04%)
São Luís	12 (6.25%)	0 (0.00%)	0 (0.00%)	1 (0.52%)	82 (42.71%)	0 (0.00%)	65 (33.85%)	0 (0.00%)	0 (0.00%)	3 (1.56%)
Manaus	2 (1.04%)	0 (0.00%)	2 (1.04%)	1 (0.52%)	2 (1.04%)	2 (1.04%)	2 (1.04%)	0 (0.00%)	0 (0.00%)	4 (2.08%)
Macapá	25 (13.02%)	0 (0.00%)	5 (2.60%)	0 (0.00%)	3 (1.56%)	0 (0.00%)	3 (1.56%)	0 (0.00%)	0 (0.00%)	9 (4.69%)
Rio Branco	5 (2.60%)	0 (0.00%)	6 (3.13%)	0 (0.00%)	5 (2.60%)	6 (3.13%)	5 (2.60%)	5 (2.60%)	0 (0.00%)	2 (1.04%)

divided the process into seven distinct steps: Data Collection, Data Preprocessing, Feature Engineering, Model Selection, Model Training, Model Evaluation, and Deployment [39]. The upcoming sections provide a comprehensive breakdown of these steps in our study.

Data collection

The data collection process is a crucial step in developing ML models. It involves gathering data from various sources and ensuring the quality and integrity of the data through a well-established procedure. The quality of the final model outputs is directly influenced by the effectiveness of this step. Therefore, it is important to carefully select and prepare the data before proceeding to subsequent steps in the model development process.

As previously mentioned, the data sources used for this study comprise SINAN, which provides monthly occurrences of VL and TL cases per city from January 2007 to December 2022, and INMET, which offers monthly meteorological data (compensated mean air temperature, insolation, precipitation, and relative humidity) corresponding to the selected cities from the same period. After collecting the data, the next step involves consolidating the SINAN and INMET data to enable exploratory data analysis and the application of ML models.

Data were collected for five features in each city. Since the data is monthly and spans from January 2007 to December 2022, this results in a total of 192 records per feature for each city. Table 5 show the number and percentage of missing values and outliers for each feature across the cities. This table provides a comprehensive view of the data quality.

Data preprocessing

This stage involves employing several techniques to clean, transform, and organize the raw data, making it more suitable for the application of ML models.

At this stage, several main activities are carried out. Data cleaning involves identifying and treating missing values, outliers, and inconsistencies in the data. The transformation includes applying techniques to standardize or normalize the data to ensure uniformity. Variable Encoding encompasses transforming categorical variables into a format suitable for analysis by ML algorithms. Additionally, data balancing is typically necessary in classification tasks where the response classes are unbalanced. In such cases, an uneven distribution of classes can lead to biased model training, as the model may become overly focused on the majority class, neglecting the minority class. However, this does not apply to our situation, as we are working with a regression problem, where the goal is to predict continuous values (the number of leishmaniasis cases). In regression problems, there are no distinct classes to balance. Therefore, balancing is not required in this context, as our dataset contains numerical case counts rather than categorical class labels. In our study, we primarily focusing on data cleaning and data transformation.

Some of the independent variables obtained from INMET contained missing values. To address these gaps, an approach involving filling them in with a lag of 12, 24, 36, or 48 months was employed. This strategy is considered appropriate concerning the nature of the data. The rationale behind this approach lies in the assumption that meteorological parameters, such as precipitation for a given month, are more likely to resemble values from the same month in the previous year, two years prior, and so forth, rather than relying on the mean or median, which are the usual standard procedures. Additionally, in instances where the missing values occurred within the first 12 months, the procedure of filling them in with the subsequent value was adopted. Since the outliers represented a small proportion of the total data, no special treatment was applied to them.

Table 6
Independent variables used in the models.

Feature	Unit	Description
Insolation	Hours	Total insolation in the month
Total precipitation	Millimeter (mm)	Total precipitation in the month
Average air temperature	Degrees Celsius (°C)	Mean temperature in the month
Air humidity	Percentage (%)	Mean air humidity in the month
Leishmaniasis	Not applicable	Number of cases in the month
lag_n	Not applicable	Number of cases, where $n = 1, 2, 3$ (the number of months in the past)

The data we utilized are presented on varying scales, as detailed in the appendix (Tables A.1 to A.6). Therefore, we normalized them to a scale from 0 to 1. This was achieved by subtracting the minimum value within each column from the respective values and subsequently dividing by the difference between the column's maximum and minimum values. To simplify this process, the MinMaxScaler from the Scikit-learn [40] library was employed.

Feature engineering

This step focuses on creating and modifying features used to train the models. Some of the main procedures performed during Feature Engineering include Feature Creation, which consists of developing new features based on existing features, and Feature Selection, which chooses the most relevant and informative features for the task at hand. This process reduces the dimensionality of the data and improves computational efficiency.

The prediction of infectious disease outbreaks can be influenced by the number of past cases, as observed in several studies, including Aleixo et al. [16], Jain et al. [15], and Chumpu et al. [19]. To account for this, three additional independent variables were generated based on the number of cases. These variables were labeled as lag1, lag2, and lag3, and correspond to the case quantity with delays of 1, 2, or 3 months respectively. In addition to the meteorological variables from INMET, three more variables were added, bringing the total number of independent variables to seven, as detailed in Table 6.

The correlation of each variable with the number of cases was verified, except for the variables related to delayed quantity. It was observed that for the cities that would be for VL, insolation showed a positive correlation in all of them. Additionally, in Teresina and São Luís, the compensated mean air temperature also demonstrated the same pattern. For the cities designated for TL, the total precipitation variable had a positive correlation in all cities. Furthermore, in Manaus and Macapá, relative humidity was also positively correlated. In Rio Branco, both insolation and compensated mean air temperature were also positively correlated.

As our aim was to extract the best model among all those tested, we chose to combine all the features at hand. After the Feature Creation stage, we had seven features. Therefore, the total number of combinations is $2^n - 1 = 2^7 - 1 = 127$. While the formula for grouping all possibilities would be 2^n, it includes the empty set, which does not apply to our situation. Thus, this empty set was omitted, while all the other combinations were tested in the models.

Model selection

For infectious diseases, such as dengue, a variety of ML-based algorithms, including ANN, have been used to predict outbreak, as observed in the literature review. However, in the context of leishmaniasis, the predominant approach has traditionally been the application of time series models.

The initial model used was an ANN known as a Simple Feedforward Neural Network (SFNN). The specific architecture is a sequential neural network comprising dense layers, i.e. fully connected layers. Within this network, there is an input layer and a single hidden layer structured as a dense layer, where each neuron in the previous layer connects to every neuron in the current layer. Following the hidden layer, a dropout layer is introduced to mitigate overfitting during the training phase, where 20% of the neurons in this layer are randomly deactivated. The network concludes with a dense output layer, where all neurons from the preceding layer connect to the singular neuron in this layer. This design features a single neuron due to the regression nature of the problem, making the linear activation function the most suitable. For SFNN, an SFNN1 was initially created, which was utilized in the grid search with either 4 or 8 neurons. Subsequently, a variation called SFNN2 was developed, employing 12, 16, 20, 24, and 28 neurons for the grid search, in addition to the other hyperparameters common to both models.

The second ANN tested was the Long Short-Term Memory (LSTM) recurrent neural network. This model comprises an input layer, an LSTM layer, and a dropout layer with a dropout rate of 0.2, implemented to prevent overfitting. The model is further complemented by a dense layer consisting of one unit with a linear activation function, serving as the output layer.

Continuing from the Simple Feedforward Neural Network, an additional dense hidden layer and a dropout layer were incorporated, thereby creating the Deep Feedforward Neural Network (DFNN). The final structure includes an input layer, a dense hidden layer, a dropout layer, another dense hidden layer, another dropout layer, and, finally, an output layer.

Keras was used to build these ANN models. Keras is the high-level API of TensorFlow, an open-source platform developed by Google for numerical computation and machine learning [41].

Finally, the Support Vector Regressor (SVR) is the model applied and serves as one of the references for performance comparison, as highlighted in Roster and Rodrigues [14]. The SVR algorithm was used from the scikit-learn library, with a set of hyperparameters (kernel, gamma, C) optimized through grid search to identify the most effective combination. The kernel, gamma, and C parameters in SVR are essential for controlling the model's complexity, the influence of individual data points, and the balance between the model's flexibility and its generalizability [40].

A diagram illustrating the models utilized is presented in Fig. 6.

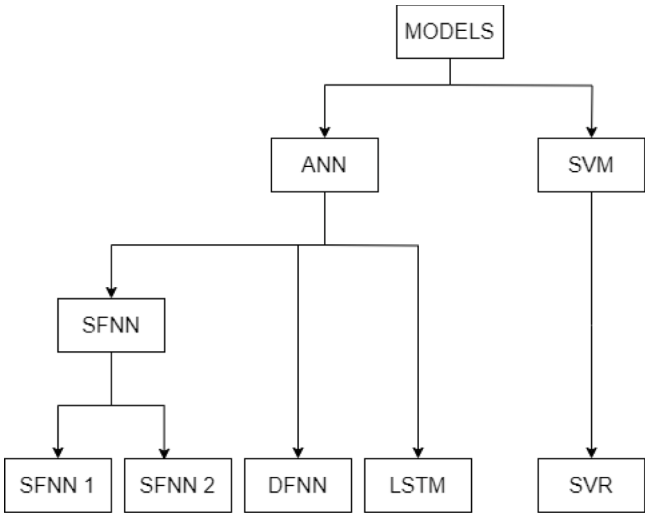


Fig. 6. Machine learning models in our study.

Model training

As previously mentioned, there are seven independent variables: compensated mean air temperature, insolation, precipitation, relative humidity, lag1, lag2, and lag3. Additionally, there is one dependent variable, which is the total number of leishmaniasis cases, identified by column "leishmaniasis". The data is divided into input and output variables, resulting in one data frame containing variables excluding the "leishmaniasis" column, which serves as the input variables, and another data frame containing only the information from the "leishmaniasis" column, representing the output variable.

The input and output variables are partitioned into training and testing sets. The training set size was determined as 81.25% of the total dataset, comprising data from January 2007 to December 2019. The test set size was calculated as the complement of the training set size (18.75%), with data from January 2020 to December 2022, aiming to evaluate the models with the statistical metrics.

The GridSearchCV function from the scikit-learn library was employed to conduct a grid search aimed at identifying the best hyperparameters for each model. This function carries out cross-validation during the grid search process, where it partitions the dataset into folds to train the model using different hyperparameter combinations, assessing the model's performance in each fold. The evaluation metric used for assessing model performance across different combinations of hyperparameters during grid search was the Mean Squared Error (MSE). Lower values of this error are indicative of superior performance.

The cross-validation method utilized in the Grid-search function involves partitioning the data into five approximately equal-sized folds. Throughout cross-validation, GridSearchCV trains and evaluates the model five times, once for each fold. In each iteration, one fold is set aside as the validation set while the remaining four folds constitute the training set.

During each iteration, the MSE metric is computed using the model's predictions on the validation data and the corresponding actual values. Following the five iterations of cross-validation, GridSearchCV computes the average of the evaluation metrics obtained across all folds. This average serves as the final model score for the specific combination of hyperparameters under evaluation. After testing all the specified hyperparameter combinations, GridSearchCV selects the model that achieved the best average score during cross-validation.

At the end of the training phase, the model is expected to have learned several patterns from the dataset, enabling it to make predictions relative to new data ("unseen data"). The final performance of the model is evaluated on test datasets to ensure its generalization ability.

A computer with CPU Intel I9-10850K @ 3.60 GHz, 128 GB of RAM, Windows 10 Pro x64, was utilized to run the models. All runs were on CPU. On average, the SFNN1, LSTM, and SVR training required approximately 21 h per city. The SFNN2 training lasted around 16 h each, while DFNN training took about 24 h for each city. The hyperparameters used for all ANN are shown in Table 7, and Table 8 are for SVR.

Table 7
Hyperparameters in artificial neural networks.

Model	Parameter	Values	Description
SFNN1/SFNN2/ DFNN/LSTM	Epochs	100, 300	Number of epochs to train the model
SFNN1/SFNN2/ DFNN/LSTM	Optimizer	Adam, RMSprop	Optimization algorithm used to update the weights
SFNN1/SFNN2/ DFNN/LSTM	Activation	Relu, Tanh	Mathematical function applied to the output of a neuron
SFNN1/ LSTM	Neurons	4, 8	Basic computational units
SFNN2	Neurons	12, 16, 20, 24, 28	Basic computational units
DFNN	Neurons	4, 8, 12, 16, 20, 24, 28	Basic computational units

Table 8
Hyperparameters in support vector regressor.

Parameter	Values	Description
kernel	Linear, poly, rbf, sigmoid	Transform the data into a higher-dimensional space
gamma	Scale, auto	Kernel coefficient
C	0.1, 1, 10, 100	Controls the trade-off between maximizing the margin and minimizing the classification error

Model evaluation

This step is critical in ML model development. The trained model is subjected to evaluation using the test dataset, which was not used during the training phase. This provides an assessment of the model's performance ability. Depending on the nature of the problem, several evaluation metrics can be used. Common metrics include precision, recall, F1-score, and confusion matrix for classification tasks. For regression problems, metrics such as Mean Absolute Error (MAE) and MSE are often used. They are respectively presented in Eq. (1) and (2), where n is the total number of observations, $\hat{y}_i$ is the predicted value, and y_i is the actual value observed for moment i [42].

The MSE was employed as the loss function to train and optimize the ANN models. This meant that during training, the optimization algorithm strove to minimize the MSE between the model prediction and the actual values of the training set. A grid search with cross-validation was utilized to find the best hyperparameters for the model. During this search, MSE was used as the scorer, testing different combinations of hyperparameters, evaluating the performance of each combination using MSE, and selecting the one that resulted in the lowest MSE.

The evaluation metrics selected to be collected are MAE, which is the average of the absolute differences between model predictions and actual values (1); MSE, which is the average of squared differences between model predictions and actual values (2); and RMSE, which is the square root of the MSE (3) [42]. For the final evaluation of the models, we used RMSE because it penalizes larger errors more than the MAE metric, making it more sensitive to outliers. In addition, it is interpreted on the same scale as the original data, which facilitates understanding.

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{1}$$

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{2}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{3}$$

The dataset combinations were tested and their results were analyzed. The combination that showed the best performance for each of the metrics was extracted, with all the information related to the variables and hyperparameters.

Table 9
Results for Fortaleza – visceral leishmaniasis.

Model	RMSE	MSE	MAE
SFNN1	3.893	15.156	3.252
SFNN2	4.181	17.480	3.519
DFNN	4.204	17.671	3.448
LSTM	6.198	38.419	5.178
SVR	**3.712**	13.782	3.179

Table 10
Results for Teresina – visceral leishmaniasis.

Model	RMSE	MSE	MAE
SFNN1	4.138	17.125	3.407
SFNN2	3.986	15.892	3.234
DFNN	**3.824**	14.622	3.195
LSTM	4.578	20.957	3.954
SVR	4.148	17.206	3.467

Deployment

In this phase, the model is computationally implemented into a system or platform that enables it to make forecasts or decisions based on new input data. Additionally, the model is made available to end users, applications, or systems that can benefit from its predictive capabilities. After evaluating the performance of all models, we selected the one for each city with a specific set of hyperparameters and variables as input. With this selection, we were able to save the model and make it available for new predictions.

4. Results and discussions

In evaluating the models, the results of the three metrics were systematically collected, as mentioned earlier. However, the decision regarding the one to be preserved and made available was based on the RMSE metric. The models were trained sequentially, starting from SFNN1, LSTM, and SVR. Based on these results, we increased the number of neurons in the Simple Feedforward Neural Network, resulting in SFNN2. In our endeavor to find improved models, we introduced DFNN, which stands for Deep Feedforward Neural Network.

Initially, we demonstrated the results of the selected cities for VL, which are Fortaleza, Teresina, and São Luís. In Fortaleza, the SVR model demonstrated the best performance using a combination of all seven input variables, resulting in an RMSE of 3.712, as shown in Table 9. Attempts to enhance performance through modifications of the ANN architecture did not surpass this RMSE value.

The DFNN model was the best for Teresina in all three metrics. The input variables were "insolation", "lag2" and "lag3". The SFNN2 also performed well using the input variables "insolation", "total_precipitation", "relative_humidity", "lag2", "lag3" (Table 10).

The assessment for São Luís showed that the LSTM model performed slightly better than other models, achieving an RMSE of 3.855. This

Table 11
Results for São Luís – visceral leishmaniasis.

Model	RMSE	MSE	MAE
SFNN1	3.916	15.337	2.982
SFNN2	3.951	15.609	2.968
DFNN	3.933	15.466	2.925
LSTM	**3.855**	14.861	2.818
SVR	3.888	15.120	3.145

Table 12
Results for Manaus – tegumentary leishmaniasis.

Model	RMSE	MSE	MAE
SFNN1	17.430	303.803	14.534
SFNN2	**16.911**	285.966	13.921
DFNN	17.264	298.050	14.408
LSTM	22.414	502.397	19.564
SVR	17.614	310.239	14.578

Table 13
Results for Rio Branco – tegumentary leishmaniasis.

Model	RMSE	MSE	MAE
SFNN1	**6.056**	36.673	4.913
SFNN2	6.424	41.262	5.475
DFNN	6.084	37.017	4.817
LSTM	6.326	40.015	4.698
SVR	7.033	49.459	6.067

Table 14
Results for Macapá – tegumentary leishmaniasis.

Model	RMSE	MSE	MAE
SFNN1	4.851	23.528	3.974
SFNN2	4.847	23.489	4.043
DFNN	**4.778**	22.830	3.964
LSTM	5.017	25.170	4.139
SVR	4.850	23.523	3.978

LSTM model used features such as “insolation”, “temperature”, “humidity”, “lag1”, and “lag3”. Despite trying other combinations of input variables and neural network configurations afterward, none surpassed this performance, as shown in Table 11.

The results of the selected cities for TL are presented ahead, which are Manaus, Rio Branco, and Macapá. For Manaus, after testing all conceivable combinations of input features, the best initial model was an SFNN1 with an RMSE of 17.43, relying solely on “lag1”. The SFNN2 slightly improved the performance, yielding an RMSE of 16.911 using the same feature, “lag1”. The best models for Manaus did not incorporate meteorological variables. All performance metrics are detailed in Table 12.

Concerning Rio Branco, the ANN models performed fairly well and the RMSE was relatively close. The SVR model performed a little worse. The SFNN1 with “insolation” and “lag1” variables achieved the lowest RMSE of 6.056, Table 13.

In Macapá, the SFNN1 and SVR models exhibited a similar performance, both achieving an RMSE of 4.85, although they utilized different combinations of input variables. In SFNN1, they were “insolation”, “lag1”, and “lag2”. In SVR, they were “insolation”, “humidity”, “lag1”, and “lag3”. The DFNN with “humidity”, “lag1”, “lag2”, and “lag3” achieved the best performance, yielding an RMSE of 4.778, Table 14.

Table 15 presents the final results of the best-performing models, with the selected variables, fine-tuned hyperparameters, and the evaluation metric used to assess their performance. After selecting the best models for the cities, we applied them to the test data. In the following graphs, the X-axis represents the months spanning from January 2020 to December 2022, while the Y-axis depicts the total number of cases. The orange line corresponds to the predictions, and the blue line represents the true values.

It is not appropriate to use RMSE solely to compare models among the cities as they have different scales in the number of cases. In the case of Manaus, the model demonstrates higher RMSE values compared to other cities (Table 12); however, graphical analysis indicates that it tracks real values more accurately, as Fig. 7, despite not using meteorological data. It is important to note that Manaus has a different scale, exceeding 80 total cases, while other cities range from 0 to around 30.

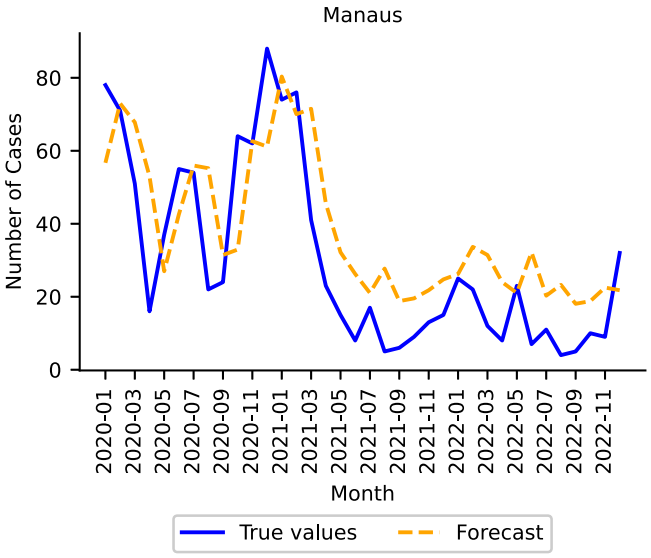


Fig. 7. Predict chart for Manaus.

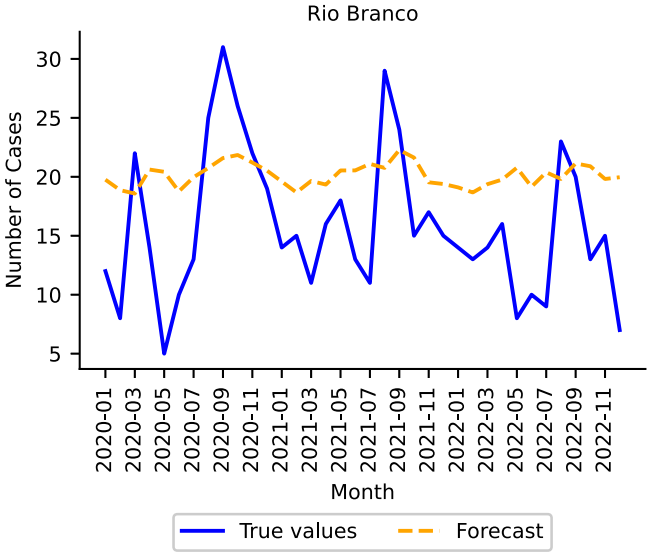


Fig. 8. Predict chart for Rio Branco.

Despite having RMSE values close to Macapá, graphically, we can see that the model does not fit the data well for Rio Branco, as shown in Fig. 8 compared to Fig. 9. Therefore, new approaches are required, such as the inclusion of new variables or even trying different models. Indeed, we tried other models (e.g., Random Forest, Gradient Boosting, XGBoost, and LightGBM), but they all yielded worse results compared to the current ones.

The same is observed for the cities studied for VL. Despite exhibiting similar metrics, they did not perform similarly on test data. The graphical analysis reveals that the model for Fortaleza in Fig. 10 outperformed the predictions compared to Teresina in Fig. 11 and São Luís in Fig. 12.

As observed in the figures above, for the cities of Fortaleza, Manaus, and Macapá, the models learned the pattern and made accurate predictions in the test data. However, for Teresina, São Luís, and Rio Branco, the predictions were not as accurate; Rio Branco performed particularly poorly.

Table 15
Overview of the key features in the top performing models.

City	Model	Input variables	Hyperparameters	RMSE
Fortaleza	SVR	Insolation, precipitation, air temperature, air humidity, lag1, lag2, lag3	C: 10, gamma: scale, kernel: linear	3.712
Teresina	DFNN	Insolation, lag2, lag3	Activation: relu, epochs: 100, neurons: 20, optimizer: Adam	3.824
São Luís	LSTM	Insolation, air temperature, air humidity, lag1, lag3	Activation: relu, epochs: 100, neurons: 8, optimizer: RMSprop	3.855
Manaus	SFNN2	Lag1	Activation: tanh, epochs: 300, neurons: 16, optimizer: RMSprop	16.911
Rio Branco	SFNN1	Insolation, lag1	Activation: tanh, epochs: 100, neurons: 8, optimizer: Adam	6.056
Macapá	DFNN	Air humidity, lag1, lag2, lag3	Activation: tanh, epochs: 300, neurons: 12, optimizer: RMSprop	4.778

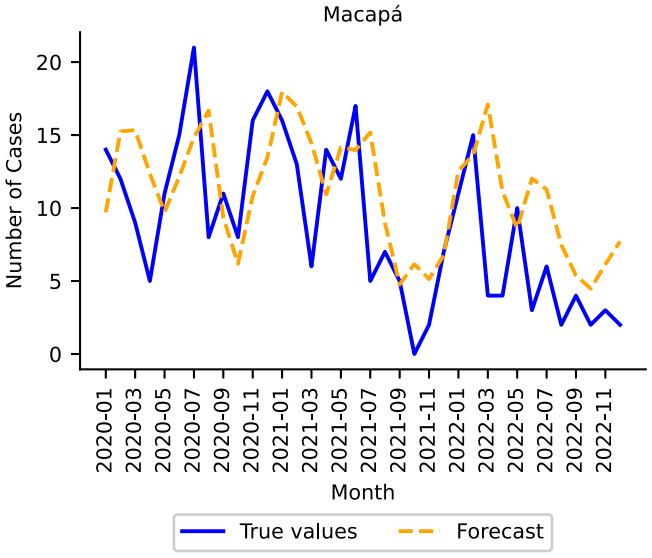


Fig. 9. Predict chart for Macapá.

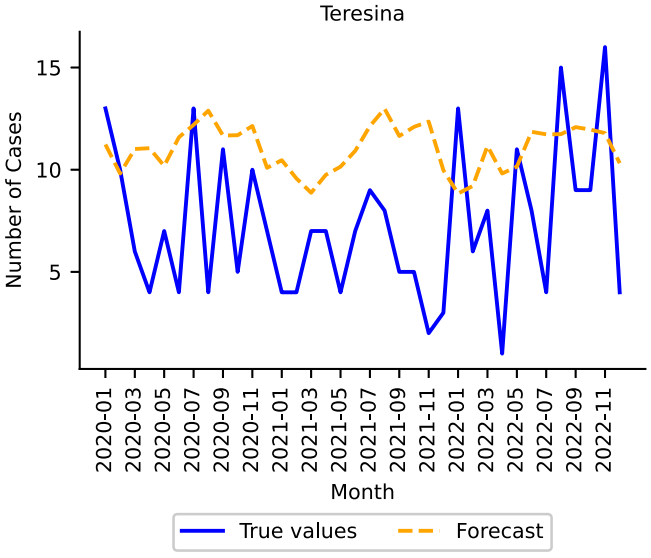


Fig. 11. Predict chart for Teresina.

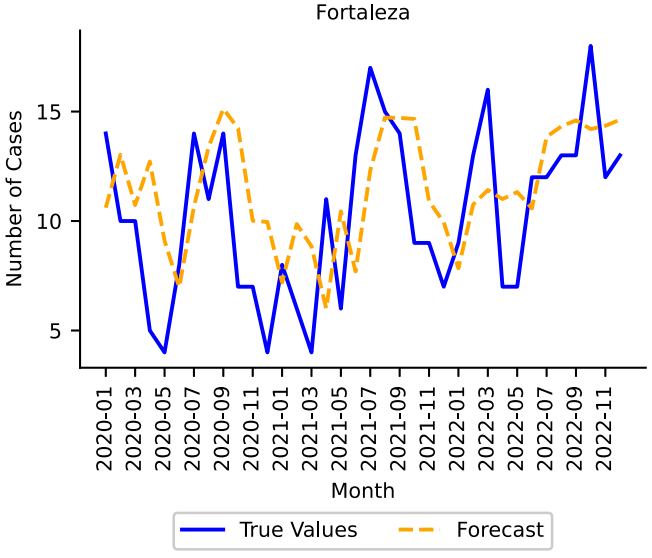


Fig. 10. Predict chart for Fortaleza.

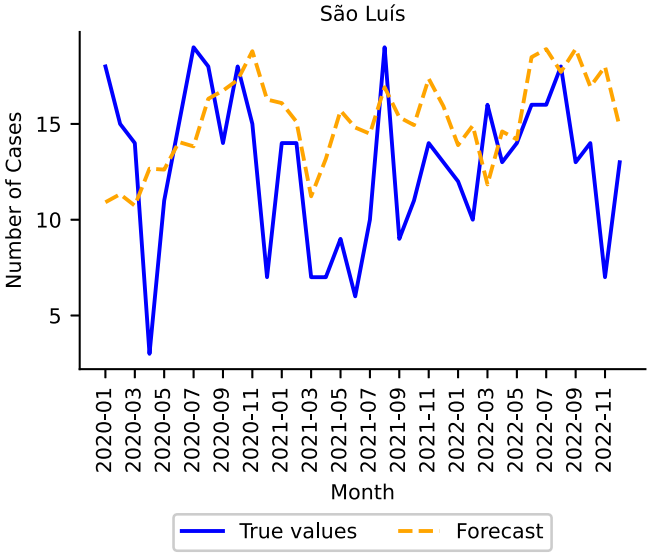


Fig. 12. Predict chart for São Luís.

5. Concluding remarks

In this paper we explored the novel application of machine learning models in predicting leishmaniasis outbreaks. Both forms of the disease were considered, the visceral form and the tegumentary, which can be found in Brazil. The models include a Simple Feedforward Neural Network, a Deep Feedforward Neural Network, a Long Short-Term Memory Recurrent Neural Network, and a Support Vector Regressor algorithm. The study focused on the three cities with the highest incidence of each form of the disease from 2007 to 2022, for which meteorological data was available for correlation within the models. The selected cities for visceral leishmaniasis were Fortaleza, Teresina, and São Luís, while for tegumentary leishmaniasis, they were Manaus, Rio Branco, and Macapá.

The models incorporated meteorological variables as inputs, such as insolation, compensated mean air temperature, total precipitation, and relative humidity. Moreover, three lag variables were created to account for disease case delays of 1, 2, and 3 months. The models were evaluated using various metrics, in which the Root Mean Squared Error (RMSE) served as the key criterion for selecting the final model.

Several strategies were employed to enhance the models' performance, including a loop to explore different feature sets and test all possible hyperparameter combinations. The data were divided into training and testing sets, and a cross-validation strategy was used to achieve optimal results.

Although the models were tailored to each city's specific data, they can still be applied in other cities. In cases where creating city-specific models is not feasible, such as when data is lacking, these models can be used to generalize leishmaniasis outbreak forecasts. A significant finding emerged: insolation and case lag were consistent factors in all models for visceral leishmaniasis, while case lag was the only consistent factor across all models for tegumentary leishmaniasis.

In spite of the relatively low RMSE exhibited by the cities of Teresina (3.824), São Luís (3.855), and Rio Branco (6.056) when predicting leishmaniasis outbreaks, the results were less favorable when examined through graphical analysis. This discrepancy indicates that while the models provided accurate numerical estimates, they may have failed to capture complex patterns in the data or to generalize effectively to new observations. This mismatch between numerical performance metrics and practical effectiveness underscores the importance of a more comprehensive evaluation of machine learning models, encompassing not only statistical measures but also detailed visual analysis, to ensure that predictions are useful and reliable in practice.

It is interesting to note that for the cities of Fortaleza, Manaus, and Macapá, with RMSE values of 3.712, 16.911, and 4.778, respectively, the graphical analysis confirmed the models' satisfactory performance. This suggests that the models were effective in capturing data patterns and generalizing well to new cases.

The models presented here provide new opportunities for public health planning, enabling authorities to implement targeted interventions, allocate resources efficiently, and enhance overall disease control strategies. Continuous monitoring of leishmaniasis cases, combined with predictive capabilities, serves as a valuable additional tool.

In addition to efforts to predict outbreaks, society can take actions to help reduce the number of infected individuals. These measures include proper waste disposal, maintaining the environment clean, avoiding debris accumulation that can foster mosquito breeding, using repellents, wearing long-sleeved clothing and pants in high-risk areas, installing protective barriers on windows and doors, and using mosquito nets on beds.

Domestic animals such as dogs and cats are considered accidental hosts of leishmaniasis, acting as protozoan reservoirs for insects that can transmit the disease to humans. Since 1963, Brazil has required mandatory reporting of leishmaniasis cases in animals. In 2021 a new law came into force in the country, stipulating that animal euthanasia is permitted only for incurable cases that pose a risk to human health or the health of other animals, whereas previously the procedure was mandatory for leishmaniasis. Laws regarding how to address the disease remain a topic of social debate.

To prevent contamination, protective measures such as insecticide-treated collars, mosquito nets, and animal vaccinations can be employed. In Brazil, the only vaccine available for dogs is Leish-Tec, which shows promise but needs further improvements to achieve full effectiveness in real-world conditions.

Limitations of the study include the lack of meteorological data for cities with the highest incidence of the disease in Brazil. Access to this data would be valuable, allowing for an analysis of the most affected cities and presenting more targeted insights. Additionally, the absence of information regarding infected animals, particularly dogs, which act as accidental hosts for the parasite, is a notable limitation. Including such data could potentially enhance the accuracy of predictive models.

The non-utilization of socioeconomic data in the models could also be a limitation, as factors such as living conditions, access to health services, and infrastructure may influence the incidence of the disease. Incorporating these variables could enrich the analysis and offer a more comprehensive understanding of the determinants of leishmaniasis.

Future research in this area could involve expanding the models to include more cities, exploring new models and input variables, and experimenting with a wider range of hyperparameter combinations. From a computational standpoint, using Graphic Processing Units instead of Central Processing Units can potentially speed up processing and deliver results more quickly. Another direction could combine the prediction outbreaks with optimization models, e.g., by determining the assignment of health resources to the cities according to the forecasts or carrying out preventive actions by the district authorities to control the proliferation of mosquitoes.

CRediT authorship contribution statement

André Cintas Donizette: Methodology, Software, Validation, Formal analysis, Investigation, Data Curation, Writing – original draft, Writing – review & editing, Visualization. **Cleber Damião Rocco:** Conceptualization, Methodology, Validation, Writing – review & editing, Supervision, Funding acquisition. **Thiago Alves de Queiroz:** Conceptualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We acknowledge the National System of Information on Notifiable Diseases (SINAN) for providing the leishmaniasis data and the National Weather Institute (INMET) for the meteorological data. This study received financial support from the National Council for Scientific and Technological Development, Brazil (CNPq: 402669/2023-1, 408722/2023-1, and 315555/2023-8), the Fundação de Desenvolvimento da Unicamp, Brazil (Funcamp, Faepex PIND: 2533/23), and the State of São Paulo Research Foundation, Brazil (FAPESP: 23/04285-1). Finally, we thank the editor-in-chief for processing our article, the two reviewers for their valuable suggestions, and the journal manager for the careful editing.

Appendix. Descriptive analysis

This appendix presents a descriptive analysis of the data for each city in Brazil selected in this study (see Tables A.1–A.6).

Table A.1
Descriptive data analysis for Fortaleza.

	Insolation	Total precipitation	Average air temperature	Relative air humidity	Leishmaniasis cases
count	192	192	192	192	192
mean	228.5	127.652	27.217	77.951	19.505
std	55.086	139.037	0.592	5.07	9.307
min	98	0	25.728	68.183	4
25%	185.65	10.675	26.781	73.452	12
50%	237.25	73.4	27.234	77.613	18
75%	277.8	211.925	27.648	82.211	25
max	311.3	668.3	28.763	90.177	49

Table A.2
Descriptive data analysis for Teresina.

	Insolation	Total precipitation	Average air temperature	Relative air humidity	Leishmaniasis cases
count	192	192	192	170	192
mean	235.153	112.176	27.61	70.508	12.536
std	54.021	125.377	1.216	12.022	5.437
min	112.2	0	25.441	45.692	1
25%	195.95	10.75	26.954	61.055	9
50%	240.45	61.4	27.332	74.417	12
75%	281.35	187.6	27.937	80.778	16
max	326.5	546.5	30.977	89.355	29

Table A.3
Descriptive data analysis for São Luís.

	Insolation	Total precipitation	Average air temperature	Relative air humidity	Leishmaniasis cases
count	192	192	192	192	192
mean	192.316	180.449	27.57	81.339	14.443
std	62.969	194.781	0.677	4.517	8.405
min	59.2	0	25.519	74.218	0
25%	137	11.525	27.163	78.403	7.75
50%	198.9	109.85	28.023	78.403	13
75%	247.375	317.45	28.103	84.952	19
max	293.4	818.2	28.455	92.589	45

Table A.4
Descriptive data analysis for Manaus.

	Insolation	Total precipitation	Average air temperature	Relative air humidity	Leishmaniasis cases
count	192	192	192	192	192
mean	192.316	180.449	27.570	81.339	14.443
std	62.969	194.781	0.677	4.517	8.405
min	59.2	0	25.519	74.218	0
25%	137	11.525	27.163	78.403	7.75
50%	198.9	109.85	28.023	78.403	13
75%	247.375	317.45	28.103	84.952	19
max	293.4	818.2	28.455	92.589	45

Table A.5
Descriptive data analysis for Macapá.

	Insolation	Total precipitation	Average air temperature	Relative air humidity	Leishmaniasis cases
count	192	192	192	192	192
mean	205.586	213.598	27.571	79.721	11.781
std	67.492	155.266	0.988	6.471	10.212
min	70.2	0	25.817	66.758	0
25%	146.7	78.175	26.751	74.373	5
50%	208.75	208.35	27.483	81.448	8
75%	266.975	328	28.401	85.334	16
max	307.6	584.5	29.447	88.815	54

Table A.6
Descriptive data analysis for Rio Branco.

	Insolation	Total precipitation	Average air temperature	Relative air humidity	Leishmaniasis cases
count	192	192	192	192	192
mean	166.961	168.877	25.610	84.416	20.542
std	48.700	127.617	1.037	5.642	7.360
min	76.2	0	22.084	67.25	2
25%	129.2	51.3	24.961	81.425	15
50%	162.9	150.7	25.796	86.125	20
75%	203.325	253.05	26.284	88.141	25
max	280.5	512.2	27.742	94.468	42

Data availability

Data and codes are available in the Github repository: https://github.com/AndreCintas/leishmaniasis.

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